Gaussian processess

- Kernel regression
- Definition

Kernel regression

We have data $\left\{(x_n^\sharp, y_n^\sharp, \epsilon_n^\sharp)\right\}_{n=1}^{N^\sharp} \subset \mathbb{X} \times \mathbb{R} \times \mathbb{R}$ and seek predictions at $\left\{x_n^\star\right\}_{n=1}^{N^\star} \subset \mathbb{X}$ of an unknown function

$$f : \mathbb{X} \mapsto \mathbb{R}$$

We start with a mean and a symmetric positive semi-definite kernel

$$m : \mathbb{X} \mapsto \mathbb{R}$$
$$v : \mathbb{X} \times \mathbb{X} \mapsto \mathbb{R}$$

In kernel regression, we follow these steps:

1. We discretize our input space
2. We discretize our regressor
3. We discretize our prior mean and covariance
4. We form our Bayesian model
5. We obtain our marginal posterior

Kernel regression

- Step 1: we discretize our input space

$$x^\sharp = \begin{pmatrix} x_1^\sharp \\ \vdots \\ x_{N^\sharp}^\sharp \end{pmatrix} \in \mathbb{X}^{N^\sharp} \qquad x^\star = \begin{pmatrix} x_1^\star \\ \vdots \\ x_{N^\star}^\star \end{pmatrix} \in \mathbb{X}^{N^\star}$$

- Step 2: we discretize our regressor

$$f^\sharp = \begin{pmatrix} f(x_1^\sharp) \\ \vdots \\ f(x_{N^\sharp}^\sharp) \end{pmatrix} \in \mathbb{R}^{N^\sharp} \qquad f^\star = \begin{pmatrix} f(x_1^\star) \\ \vdots \\ f(x_{N^\star}^\star) \end{pmatrix} \in \mathbb{R}^{N^\star}$$

Kernel regression

- Step 3: we discretize our prior mean...

$$m^\sharp = \begin{pmatrix} m(x_1^\sharp) \\ \vdots \\ m(x_{N^\sharp}^\sharp) \end{pmatrix} \in \mathbb{R}^{N^\sharp} \qquad m^\star = \begin{pmatrix} m(x_1^\star) \\ \vdots \\ m(x_{N^\star}^\star) \end{pmatrix} \in \mathbb{R}^{N^\star}$$

...and our prior covariance

$$v^{\sharp\sharp} = \begin{bmatrix} v(x_1^\sharp, x_1^\sharp) & \cdots & v(x_1^\sharp, x_{N^\sharp}^\sharp) \\ \vdots & \ddots & \vdots \\ v(x_{N^\sharp}^\sharp, x_1^\sharp) & \cdots & v(x_{N^\sharp}^\sharp, x_{N^\sharp}^\sharp) \end{bmatrix} \in \mathbb{R}^{N^\sharp \times N^\sharp}_{sym,+} \qquad v^{\sharp\star} = \begin{bmatrix} v(x_1^\sharp, x_1^\star) & \cdots & v(x_1^\sharp, x_{N^\star}^\star) \\ \vdots & \ddots & \vdots \\ v(x_{N^\sharp}^\sharp, x_1^\star) & \cdots & v(x_{N^\sharp}^\sharp, x_{N^\star}^\star) \end{bmatrix} \in \mathbb{R}^{N^\sharp \times N^\star}$$

$$v^{\star\sharp} = \begin{bmatrix} v(x_1^\star, x_1^\sharp) & \cdots & v(x_1^\star, x_{N^\sharp}^\sharp) \\ \vdots & \ddots & \vdots \\ v(x_{N^\star}^\star, x_1^\sharp) & \cdots & v(x_{N^\star}^\star, x_{N^\sharp}^\sharp) \end{bmatrix} \in \mathbb{R}^{N^\star \times N^\sharp} \qquad v^{\star\star} = \begin{bmatrix} v(x_1^\star, x_1^\star) & \cdots & v(x_1^\star, x_{N^\star}^\star) \\ \vdots & \ddots & \vdots \\ v(x_{N^\star}^\star, x_1^\star) & \cdots & v(x_{N^\star}^\star, x_{N^\star}^\star) \end{bmatrix} \in \mathbb{R}^{N^\star \times N^\star}_{sym,+}$$

Kernel regression

- Step 4: we form our Bayesian model

$$\begin{pmatrix} f^\sharp \\ f^\star \end{pmatrix} \sim \text{Normal}_{N^\sharp + N^\star} \left(\begin{pmatrix} m^\sharp \\ m^\star \end{pmatrix}, \begin{bmatrix} v^{\sharp\sharp} & v^{\sharp\star} \\ v^{\star\sharp} & v^{\star\star} \end{bmatrix} \right)$$
$$y^\sharp \,\Big|\, f^\sharp \sim \text{Normal}_{N^\sharp} \left(f^\sharp, H^{\sharp\sharp} \right)$$

Here, we gather measurements and errors in

$$y^\sharp = \begin{pmatrix} y_1^\sharp \\ \vdots \\ y_{N^\sharp}^\sharp \end{pmatrix} \in \mathbb{R}^{N^\sharp} \qquad H^{\sharp\sharp} = \begin{bmatrix} (\epsilon_1^\sharp)^2 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & (\epsilon_{N^\sharp}^\sharp)^2 \end{bmatrix} \in \mathbb{R}^{N^\sharp \times N^\sharp}_{sym,+}$$

- Step 5: we obtain our marginal posterior

$$f^\star | y^\sharp \sim \text{Normal}_{N^\star} \left(\bar{m}^\star, \bar{v}^{\star\star} \right)$$

...steps 1-5 define a recipe to work with probability distributions over uncertain functions

...these are non-parametric distributions

Gaussian processes

A shorthand like

$$f \sim \mathrm{GP}_{\mathbb{X}}(m, v)$$

denotes a Gaussian process over $\mathbb{X}$

A Gaussian process is a random function

$$f : \mathbb{X} \mapsto \mathbb{R}$$

The parameters in the Gaussian process prior are

- the mean function m which is a function

$$m : \mathbb{X} \mapsto \mathbb{R}$$

- the covariance kernel v which is also a function

$$v : \mathbb{X} \times \mathbb{X} \mapsto \mathbb{R}$$

  the covariance kernel is symmetric and positive semi-definite

Gaussian processes

$$\overbrace{f}^{\text{Gauss. proc.}} \sim \overbrace{\mathrm{GP}_{\mathbb{X}}(m, v)}^{\text{prior}} \qquad f : \mathbb{X} \mapsto \mathbb{R} \qquad m : \mathbb{X} \mapsto \mathbb{R} \qquad v : \mathbb{X} \times \mathbb{X} \mapsto \mathbb{R}$$

The probability distribution of a Gaussian process is given by *...there are no probability functions for GPs*

- For any finite collection of input points *...includes variation in the positioning of x_n and their sizing N*

$$x_1, x_2, \ldots, x_N \in \mathbb{X}$$

the output values

$$f(x_1), f(x_2), \ldots, f(x_N) \in \mathbb{R}$$

are random variables

- These random variables jointly follow a multivariate normal distribution

$$\begin{pmatrix} f(x_1) \\ f(x_2) \\ \vdots \\ f(x_N) \end{pmatrix} \sim \mathrm{Normal}_N \left(\begin{pmatrix} m(x_1) \\ m(x_2) \\ \vdots \\ m(x_N) \end{pmatrix}, \begin{bmatrix} v(x_1, x_1) & v(x_1, x_2) & \cdots & v(x_1, x_N) \\ v(x_2, x_1) & v(x_2, x_2) & \cdots & v(x_2, x_N) \\ \vdots & \vdots & \ddots & \vdots \\ v(x_N, x_1) & v(x_N, x_2) & \cdots & v(x_N, x_N) \end{bmatrix} \right)$$

Kernel regression revisited

We have data $\left\{(x_n^\sharp, y_n^\sharp, \epsilon_n^\sharp)\right\}_{n=1}^{N^\sharp} \subset \mathbb{X} \times \mathbb{R} \times \mathbb{R}$ of an unknown function

$$f : \mathbb{X} \mapsto \mathbb{R}$$

Kernel regression takes the form

...this fits more naturally within the Bayesian formalism

$$f \sim \mathrm{GP}_{\mathbb{X}}(m, v)$$
$$y_n^\sharp | f \sim \mathrm{Normal}\left(f(x_n^\sharp), (\epsilon_n^\sharp)^2\right), \qquad n = 1, \ldots, N^\sharp$$

for pre-selected mean and kernel functions

$$m : \mathbb{X} \mapsto \mathbb{R}$$
$$v : \mathbb{X} \times \mathbb{X} \mapsto \mathbb{R}$$

The Gaussian process prior

We consider a model $f \sim \mathrm{GP}_{\mathbb{X}}(m, v)$ and work on a problem with multiple targets of interest

...the targets of interest may include measurements, predictions, or both

- Our targets of interest are located at positions $x_1, \ldots, x_N \in \mathbb{X}$ *...our GP prior provides the statistics*

$$\begin{pmatrix} f(x_1) \\ \vdots \\ f(x_N) \end{pmatrix} \sim \mathrm{Normal}_N \left(\begin{pmatrix} m(x_1) \\ \vdots \\ m(x_N) \end{pmatrix}, \begin{bmatrix} v(x_1, x_1) & \cdots & v(x_1, x_N) \\ \vdots & \ddots & \vdots \\ v(x_N, x_1) & \cdots & v(x_N, x_N) \end{bmatrix} \right)$$

- We add another target of interest $x_? \in \mathbb{X}$ *...our GP prior provides the statistics*

$$\begin{pmatrix} f(x_1) \\ \vdots \\ f(x_N) \\ f(x_?) \end{pmatrix} \sim \mathrm{Normal}_{N+1} \left(\begin{pmatrix} m(x_1) \\ \vdots \\ m(x_N) \\ m(x_?) \end{pmatrix}, \begin{bmatrix} v(x_1, x_1) & \cdots & v(x_1, x_N) & v(x_1, x_?) \\ \vdots & \ddots & \vdots & \vdots \\ v(x_N, x_1) & \cdots & v(x_N, x_N) & v(x_N, x_?) \\ v(x_?, x_1) & \cdots & v(x_?, x_N) & v(x_?, x_?) \end{bmatrix} \right)$$

The Gaussian process prior

In compact notation, our statistics are

$$\begin{pmatrix} f_1 \\ \vdots \\ f_N \end{pmatrix} \sim \text{Normal}_N \left(\begin{pmatrix} m_1 \\ \vdots \\ m_N \end{pmatrix}, \begin{bmatrix} v_{11} & \cdots & v_{1N} \\ \vdots & \ddots & \vdots \\ v_{N1} & \cdots & v_{NN} \end{bmatrix} \right)$$

$$\begin{pmatrix} f_1 \\ \vdots \\ f_N \\ f_? \end{pmatrix} \sim \text{Normal}_{N+1} \left(\begin{pmatrix} m_1 \\ \vdots \\ m_N \\ m_? \end{pmatrix}, \begin{bmatrix} v_{11} & \cdots & v_{1N} & v_{1?} \\ \vdots & \ddots & \vdots & \vdots \\ v_{N1} & \cdots & v_{NN} & v_{N?} \\ v_{?1} & \cdots & v_{?N} & v_{??} \end{bmatrix} \right)$$

These are related by

$$p(f_1, \ldots, f_N) = \int_{-\infty}^{+\infty} df_? \, p(f_1, \ldots, f_N, f_?)$$

$$p(f_1, \ldots, f_N, f_?) = p(f_?|f_1, \ldots, f_N) \, p(f_1, \ldots, f_N)$$

$$p(f_?) = \int_{-\infty}^{+\infty} df_1 \cdots \int_{-\infty}^{+\infty} df_N p(f_1, \ldots, f_N, f_?)$$

Demonstration

We consider a Gaussian process

$$f \sim \mathrm{GP}_{\mathbb{R}}(m, v)$$

over the real line $\mathbb{R}$

- Our mean function is

$$m(x) = x^2, \qquad x \in \mathbb{R}$$

- Our covariance kernel is

$$v(x', x'') = 100 \exp\left(-\frac{(x' - x'')^2}{2}\right), \quad x', x'' \in \mathbb{R}$$

...this is a squared-exponential kernel

